Supplementary Material
Observer-Based Hybrid Position Holding for an Electrohydraulic Agricultural Hitch

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S1. Historical model and corrected interpretation

The historical pressure-difference surrogate and V2 are separate models. The old surrogate has a virtual command-to-pressure relaxation, a retained-loss multiplier and a virtual accumulator/recharge proxy. Its paired pressure outputs are reconstructed rather than separately conserved. The implemented pressure loss is $\mathcal{L} = \kappa[1 + g_T \max(T - 25, 0)] \max(d, 0)/A$. Dimensional consistency requires $[\kappa] = \mathrm{m}^2\,\mathrm{s}^{-1}$ so that $[\mathcal{L}] = \mathrm{Pa}\,\mathrm{s}^{-1}$. The old physical leakage-flow interpretation and $\mathrm{m}^3\ \mathrm{s}^{-1}\ \mathrm{Pa}^{-1}$ label are withdrawn. Old file identifiers are retained only to reproduce historical computations. The corrected coefficient-proxy inversion and a rate-dependent bound appear in Section 3.5. They do not identify physical leakage in V2.

The archived raw 0.10 MPa pressure-noise group (seeds 2060–2079) has 500 mm maximum steady error in every path, with mean steady RMSE 486.56 mm. A small reported downward drop of 0.033 mm in that group is not successful holding: the actuator travels to the upper stroke boundary. This illustrates why displacement drop alone is an insufficient acceptance metric. The V2 replay at the same absolute pressure-noise SD has 20/20 passes and no stroke hit, but its observer, reference and draw sequence also differ. This is a comparison of implementations, not a controlled attribution to chamber architecture alone.

S2. Measurement interface and original selection budget

The full observer update, pressure command, saturation, integrators, switching order, hysteresis and timers are in Sections 3.2–3.3. Position correction is 0.08, velocity correction is $0.0005/h_c$, pressure filtering is 0.18 and effective-load smoothing is 0.005 per controller sample. Initial position and pressures equal the initial plant values, velocity is zero, and load starts at 9000 N. This initialization is a virtual-study assumption; it is not evidence of physical observer convergence. The effective load is clipped to [3000,20000] N in the controller. Nominal model coefficients remain fixed when true mechanical parameters are varied.

The 27 development candidates cross dwell $\{4, 8, 12\}$ s, support-error threshold $\{2.5, 4, 6\}$ mm and support-pressure threshold $\{0.18, 0.30, 0.45\}$ MPa, evaluated for 300 s on seeds 4100–4104. All pass five development paths. Ranking is lexicographic: maximum pass count, then minimum mean oil volume, mean mode changes and mean drop. Equal ties preserve enumeration order. No revision result changes the selected settings. The complete development table follows; quantities are virtual inputs or outputs, not hardware calibration.

Table 1: Original development grid; every row passes 5/5.

Dwell (s)	Error (mm)	Deficit (MPa)	Oil (mL)	Switches	Drop (mm)
4	2.5	0.18	21.27	5.0	0.996
4	2.5	0.30	15.50	3.4	1.222
4	2.5	0.45	15.50	3.4	1.222
4	4.0	0.18	21.27	5.0	0.996
4	4.0	0.30	0.00	3.0	1.773
4	4.0	0.45	0.00	3.0	1.773
4	6.0	0.18	21.27	5.0	0.996
4	6.0	0.30	0.00	3.0	1.773
4	6.0	0.45	0.00	3.0	1.773
8	2.5	0.18	25.44	5.0	0.845
8	2.5	0.30	17.98	3.0	1.230
8	2.5	0.45	17.98	3.0	1.230
8	4.0	0.18	25.44	5.0	0.845
8	4.0	0.30	0.00	3.0	1.754
8	4.0	0.45	0.00	3.0	1.754
8	6.0	0.18	25.44	5.0	0.845
8	6.0	0.30	0.00	3.0	1.754
8	6.0	0.45	0.00	3.0	1.754
12	2.5	0.18	30.46	4.6	0.861
12	2.5	0.30	20.82	3.0	1.188
12	2.5	0.45	20.82	3.0	1.188
12	4.0	0.18	30.46	4.6	0.861
12	4.0	0.30	0.00	3.0	1.803
12	4.0	0.45	0.00	3.0	1.803
12	6.0	0.18	30.46	4.6	0.861
12	6.0	0.30	0.00	3.0	1.803
12	6.0	0.45	0.00	3.0	1.803

S3. Original frozen statistics and uncertainty

The submitted 20 paths use seeds 4120–4139, disjoint from development. Mean confidence intervals use $\bar{z} \pm t_{0.975,n-1} s/\sqrt{n}$, where s is sample SD. These are intervals for seeded model-noise means, not hardware population intervals or pass-probability guarantees. No seed is removed for failure. Drop statistics require hold entry; every submitted path enters holding. The post-transition maximum error and pass screens include failed paths.

Chain	Metric	Mean	SD	95% CI
PID–lock	Drop (mm)	16.041	0.065	16.010–16.071
PID–lock	RMSE (mm)	8.232	0.115	8.178–8.286
PID–lock	Max. error (mm)	15.138	0.132	15.076–15.200
PID–lock	Mode changes	1.000	0.000	1.000–1.000
PID–lock	Oil (mL)	0.000	0.000	0.000–0.000
SMC–lock	Drop (mm)	15.995	0.055	15.969–16.021
SMC–lock	RMSE (mm)	9.174	0.175	9.092–9.256
SMC–lock	Max. error (mm)	16.227	0.199	16.134–16.320
SMC–lock	Mode changes	1.000	0.000	1.000–1.000

Chain	Metric	Mean	SD	95% CI
SMC–lock	Oil (mL)	0.000	0.000	0.000–0.000
PID–accumulator	Drop (mm)	3.441	0.249	3.325–3.558
PID–accumulator	RMSE (mm)	0.246	0.004	0.244–0.247
PID–accumulator	Max. error (mm)	6.960	0.219	6.858–7.063
PID–accumulator	Mode changes	26.800	5.872	24.052–29.548
PID–accumulator	Oil (mL)	25.538	1.200	24.977–26.100
Observer hybrid	Drop (mm)	1.849	0.156	1.776–1.922
Observer hybrid	RMSE (mm)	1.222	0.026	1.210–1.234
Observer hybrid	Max. error (mm)	8.435	0.219	8.332–8.537
Observer hybrid	Mode changes	13.100	0.447	12.891–13.309
Observer hybrid	Oil (mL)	1.019	4.559	-1.114–3.153

S4. Integration refinement and local boundedness checks

Sensing, observer updates and commanded inputs remain on the 10 ms grid. Only the plant is substepped at 10, 5, 2.5 or 1 ms; the measurement draws are identical on common controller samples. Pressure updates use the old position/velocity and the updated spool, then the mechanical update uses the new pressures, and position uses the new velocity. This sequential explicit scheme and its pressure/stroke projections are retained. It is not an adaptive error-controlled solver. Pressure discrepancies below aggregate both chambers on all 10 ms sample endpoints against the 1 ms run. Maxima may include a switching-time displacement and are not an estimate of local truncation order.

Table 3: Full-horizon 10 ms versus 1 ms pressure and drop differences.

Seed	Pressure RMS (MPa)	Max. difference (MPa)	Drop difference (mm)
4300	0.0638	0.1510	0.1990
4301	0.0642	0.1657	0.1120
4302	0.0640	0.1583	0.2394
4303	0.0633	0.1543	-0.0326
4304	0.0633	0.1522	0.2357

The event-clipped RK4 audit keeps controller, observer and measurement updates on the 10 ms grid and changes only the virtual-plant update. Five predeclared seeds were run at 1.0 and 0.5 ms; all paths passed. The mean hold drops were 1.628 and 1.585 mm, and the mean worst post-transition errors were 8.503 and 8.502 mm. The mean absolute per-seed difference in worst error was 0.0023 mm and the mean absolute hold-drop difference was 0.046 mm. These results support fine-step numerical convergence of the refined arm, while switching and friction remain sources of hold-drop sensitivity.

Table 4: Event-clipped RK4 fine-step convergence audit.

Plant step (ms)	Pass	Mean drop (mm)	Drop SD (mm)	Mean worst error (mm)	Worst error (mm)
1.0	5/5	1.628	0.058	8.503	8.747
0.5	5/5	1.585	0.059	8.502	8.747

The projected-model velocity bound is independent of mode but depends on imposed pres-

sure/stroke limits. It is deliberately too loose for a 10 mm guarantee. For the unprojected local calculation at $\bar{x} = 0.35$ m, nominal $C_l = 5 \times 10^{-14}$ m^3 s^{-1} Pa^{-1} and 25 °C, the frozen-volume matrix in Section 3.4 has eigenvalues approximately -6.23×10^{-6} and $-0.918 \pm 73.08i$ s^{-1}. The slow pole exposes weak leakage-driven restoration rather than precise locking. The Lyapunov argument is in fixed consistent coordinates, not a physical-energy interpretation of a mixed-unit Euclidean norm. The result concerns bounded forcing and fixed volumes, with no claim that the nonlinear trajectory remains in its local neighborhood.

S5. Noise activation, storage and historical alignment

Independent Gaussian pressure noise is added before the V2 internal pressure filter; no new external low-pass filter is inserted. Position noise stays at 1.5 mm SD in the pressure-only sweep. The separate joint-noise case doubles both SDs. Dwell limits mode changes but does not prevent persistent positive support commands during a latched support mode. Pressure/position fluctuations change pressure balance and gate crossings, explaining the large change in oil volume from a near-zero nominal denominator. Gas pressure, cumulative discharge, active time and mode occupancy are recorded; energy consumption and valve wear are not inferred from these proxies.

Table 5: Five-path noise-group activity and storage extrema.

Noise input	Pass	Active time (s)	Mean oil (mL)	Min. gas (MPa)
Nominal	5/5	0.00	0.00	14.000
Both SDs doubled	5/5	6.23	39.92	13.553
Pressure SD 0.10 MPa	5/5	18.51	98.76	13.325
Pressure SD 0.20 MPa	5/5	177.44	298.04	12.204
Pressure SD 0.50 MPa	5/5	450.62	447.11	11.497
Pressure SD 1.00 MPa	2/5	479.06	454.13	11.466

The ten original jointly doubled-noise paths were replayed with diagnostic logging; all original fields match exactly. Mean support-command active time is 10.391 s (SD 5.196 s), mean oil volume 61.982 mL (SD 28.408 mL), and the minimum gas pressure across the group is 13.337 MPa. Every path passes the original screen and has no stroke hit. Figure 7 of the main article shows the first original stress seed 4160, against a same-seed nominal replay, so that the original discharge increase has a direct trajectory diagnosis rather than being explained only through a new seed set.

The aligned 20-path historical-noise replay gives mean drop 1.860 mm (95% interval 1.774–1.946), worst error 8.723 mm, and mean oil 105.36 mL. Its larger discharge is compatible with passing the position screen. Seeds and SD match the old experiment, but the stochastic samples, plant, reference and internal observer do not. An architecture-only noise-robustness causal claim is therefore withheld.

S6. Parameter provenance and revision overrides

The original 45-row parameter manifest is supplied without rewriting historical coverage labels. Every number is a researcher-defined theoretical input. The new ledger records additional true-plant overrides, integration substeps, observer-removal arms and noise scales separately. Hydraulic area, dead volume, effective bulk modulus, flow area and dead zone have not been independently identified. A declared range in the submitted table did not by itself document a performed

test. The new mechanical runs vary one true coefficient at a time and keep controller/observer inputs nominal; they cannot establish robustness over all simultaneous combinations.

S7. Reproduction and integrity

The reproduction ZIP contains current code, original JSON evidence, revision protocols/results, parameter manifest and scripts. A frozen submitted source/code copy is kept under historical submission evidence. The control defaults reproduce all original recorded fields exactly on one original path of each of the four chains; this is a regression check, not a complete numerical validation. The original 20-path JSON values are retained rather than overwritten. Run the scripts below from the reproduction root with Python 3 and NumPy, SciPy and Matplotlib. Their protocol files specify seeds before computation. The mode-scan script is historical and need not be rerun to reproduce the fixed revision evidence.

```
python3 tools/run_paper1_major_revision.py
python3 tools/extend_paper1_revision_audits.py
python3 tools/refine_paper1_comparators.py
python3 tools/replay_paper1_original_noise.py
python3 tools/plot_paper1_major_revision.py
```

The first script runs 177 primary revision simulations; the historical-noise alignment adds 20, refined comparator checks add 15, and original-noise diagnostics add 11. These are seeded theoretical replays, not 223 independent physical experiments; families deliberately reuse paired seeds. The first primary revision protocol is supplemented by the explicit mechanical-mismatch ledger. Results and source hashes are listed in the package manifest. Main article figures use predeclared revision seed 4300 or the first original noise seed 4160, not best-path selections. Public GitHub/Zenodo deposition of this revision remains pending and is not asserted as complete.

S8. Paired RK4 comparators and joint mechanical vertices

The strengthening protocol fixes all 70 rows before computation: 30 nominal comparator rows and 40 joint-vertex rows. Ten nominal hybrid rows are reused from the earlier RK4 audit, so 60 new trajectories are computed. Each path lasts 1800 s, with controller/observer/noise updates at 10 ms. The two PID arms use exactly the same five seeds and plant integrator as the hybrid; the matched arm changes only the declared latch and discharge-gating semantics. The joint screen crosses all eight $\{0.7, 1.3\}^3$ mass/damping/stiffness vertices with five seeds, a 1 ms RK4 plant step and fixed nominal control coefficients. Source hashes, per-task checkpoints and all raw outputs accompany the protocol. These vertices do not certify the full box or an uncertainty probability distribution. Endpoint projections do not locate event roots or establish a fourth-order rate for this nonsmooth system.

Table 6: Paired RK4 comparison on five common seeds per row, with frozen gains and 10 ms sensing/control. The matched PID arm adds an 8 s latch and discharge gating. RMSE is post-transition tracking error; oil is accumulator discharge.

Chain	Step (ms)	Pass	Mean drop (mm)	Mean RMSE (mm)	Mean oil (mL)	Mean changes
Observer hybrid	1.0	5/5	1.628	1.178	0.00	13.0
PID–accumulator	1.0	5/5	3.002	0.249	29.31	40.6
PID + matched latch	1.0	5/5	3.028	0.249	27.94	3.0
Observer hybrid	0.5	5/5	1.585	1.192	0.00	13.0
PID–accumulator	0.5	5/5	2.943	0.251	29.91	42.6
PID + matched latch	0.5	5/5	2.953	0.251	28.50	3.0

Table 7: Joint mechanical corner screen with the hybrid, five paired seeds per vertex and 1 ms RK4 integration. All controller/observer coefficients remain nominal. Ratios refer to true plant mass, damping and stiffness. Worst values include every path.

m/m_0	c/c_0	k/k_0	Pass	Worst drop (mm)	Worst error (mm)	Mean oil (mL)
1.3	1.3	1.3	5/5	1.595	9.155	0.00
1.3	1.3	0.7	5/5	1.756	8.352	0.00
1.3	0.7	1.3	5/5	1.583	9.023	0.00
1.3	0.7	0.7	5/5	1.692	8.284	0.00
0.7	1.3	1.3	5/5	1.746	9.153	0.00
0.7	1.3	0.7	5/5	2.009	8.351	0.00
0.7	0.7	1.3	5/5	1.750	9.177	0.00
0.7	0.7	0.7	5/5	1.965	8.372	0.00

Table 8: Paired PID-minus-hybrid differences in hold drop and RMSE with Student-t 95% intervals, five paired seeds per row. Intervals concern seeded noise at the declared plant, not uncertainty in model structure or hardware performance.

PID arm	Step (ms)	Drop difference (mm), 95% CI	RMSE difference (mm), 95% CI
Original PID	1.0	1.374 [1.149, 1.599]	-0.929 [-0.962, -0.896]
Matched PID	1.0	1.400 [1.147, 1.652]	-0.929 [-0.962, -0.896]
Original PID	0.5	1.358 [1.091, 1.625]	-0.941 [-0.966, -0.917]
Matched PID	0.5	1.368 [1.106, 1.630]	-0.941 [-0.965, -0.917]

```
python3 tools/run_paper1_rk4_convergence_audit.py
python3 tools/run_paper1_strengthening_audit.py
```